

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | ardusa.github.io | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 - Present

East Lansing, MI

- Streamlined member onboarding by automating GitHub team access and Slack workspace invitations, reducing setup time from 15 minutes to under 30 seconds.
- Built QR code event check-in system with AWS S3 bucket caching and unique token generation for automated attendance tracking.
- Established open source development workflow with GitHub Actions CI/CD, enforcing 85% code coverage; maintained comprehensive documentation and contributor guidelines to reduce development friction.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Built a full-stack note-taking application with a chapter/page model; integrated OAuth2.0 authentication with Google & GitHub providers.
- Implemented dual-database architecture (PostgreSQL + Neo4j) with compound B-tree indexes (user_id, created_at) to reduce note search latency from 850ms to 440ms.
- Designed interactive concept mapping, allowing users to connect chapters and pages with labeled arrows. ☑

PROJECTS

Mira — AI Assistant | *Next.js, Electron, FastAPI, PostgreSQL, WebHook*

May 2025 – Nov. 2025

Distributed audio analysis platform enabling intelligent background processing, and smart task execution

- Built RAG pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim) enabling context-aware automation.
- Developed RESTful FastAPI backend with lazy-loaded dependency injection for heavy ML models, optimizing for cold start time and achieving sub-900ms load on first inference.
- Combined Electron desktop client with Next.js frontend, orchestrating real-time data flow via IPC channels and a dedicated webhook server to synchronize user actions and audio streams.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation and gesture-detection

- Designed low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Created custom vector-based physics engine with AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash
Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js
Frontend Development React, Next.js, Tailwind CSS, Electron, Vite
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, PostgreSQL, Neo4j